Arnold tongue of two coupled PING circuits

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Abstract

Two independently driven PING circuits were crossed over measured natural- frequency detuning and reciprocal coupling to reproduce the synchronization result of Lowet et al. [**1**]. The locking map forms a contiguous region around zero detuning and widens across 4 successive coupling steps. All 715 primary trials are valid. The faster circuit leads in 98.7% of qualifying locked cells, but the registered rule requires the correct sign in every such cell. The reproduction therefore failed: its Arnold-tongue geometry succeeds, while one near-resolution phase cell prevents an overall pass at 80 E / 20 I. A preregistered finite-size confirmation at 800 E / 200 I subsequently passes all 40 coupled trials.

Methods

1. **Author the fixed graph.** Each circuit contains 80 conductance-based excitatory neurons and 20 inhibitory neurons. Independent 80-channel Poisson populations drive the two excitatory populations. Local E-to-I AMPA and I-to-E GABA-A projections generate PING activity. Four reciprocal AMPA projections couple E_A to E_B and I_B, and E_B to E_A and I_A. Their common weight is K. The graph and network seed remain fixed across the sweep.
2. **Calibrate uncoupled frequency.** Equal-rate, zero-coupling trials determine a monotonic operating interval from 70 to 130 Hz per input channel. Input-rate pairs target thirteen signed detunings. For each seed, the measured zero-coupling frequencies define

   $$\Delta f_0 = f_A^0 - f_B^0.$$

   Here f_A^0 and f_B^0 are the uncoupled gamma peak frequencies of circuits A and B. Δf_0, not input-rate difference or coupled frequency difference, is the horizontal coordinate.
3. **Register locking before the primary sweep.** Repeated uncoupled equal-drive trials fix the admissible emergent frequency difference, absolute relative- phase slope, and phase-slip count. A valid trial is locked only when all three estimators fall within their frozen tolerances. Phase-locking value is descriptive and does not determine the label. A bounded pilot at the two extreme detunings and zero detuning selects the coupling ceiling $K = 0.08$ before the primary map is viewed.
4. **Execute the frozen grid.** The primary design crosses thirteen target detunings with eleven coupling levels and 5 independently generated input pairs per cell. Each trajectory lasts 3000 ms at a 0.1 ms timestep; the first 500 ms are discarded. A trial is invalid if any recorded state is non-finite, any E or I firing rate is below 1 Hz, or either E population lacks a resolved 25–80 Hz spectral peak.

5. **Measure frequency and phase.** Excitatory spikes are converted to population rates with a fixed 5 ms Gaussian kernel. Gamma frequency comes from the post-transient spectrum. A zero-phase 25–90 Hz band-pass filter followed by the analytic signal gives relative phase

$$\varphi(t) = \mathrm{unwrap}\big(\theta_{A(t)} - \theta_{B(t)}\big).$$

θ_A and θ_B are the instantaneous phases of circuits A and B, and φ is their unwrapped difference. Each trial reports both frequencies, absolute frequency difference, the linear slope of φ, complete phase slips, phase-locking value, and circular mean phase.
6. **Retain reproducible evidence without dense-state bloat.** Exact input and population-spike tensors are bit-packed for every primary cell. Population- mean voltage and projection conductance are retained at 1 ms resolution, together with the compiled bundles, frozen grid, input hashes, and seed ledger. The realized runtime parameter tensors are retained once for each of the 11 coupling levels, with hashes verifying that all non-coupling tensors remain identical across levels. The simulator records dense neuron state for analysis, but the archive keeps the compact sufficient evidence rather than tens of gigabytes of redundant dense zeros.
7. **Apply the registered verdict.** The geometric criterion requires a contiguous locked region centred near zero whose width increases across at least three successive nonzero coupling levels. Every grid cell must contain a valid trial. In every locked nonzero-detuning cell, the circuit with the greater f^0 must lead in phase. The final criterion is deliberately strict: a high fraction is not substituted for the word “every”.

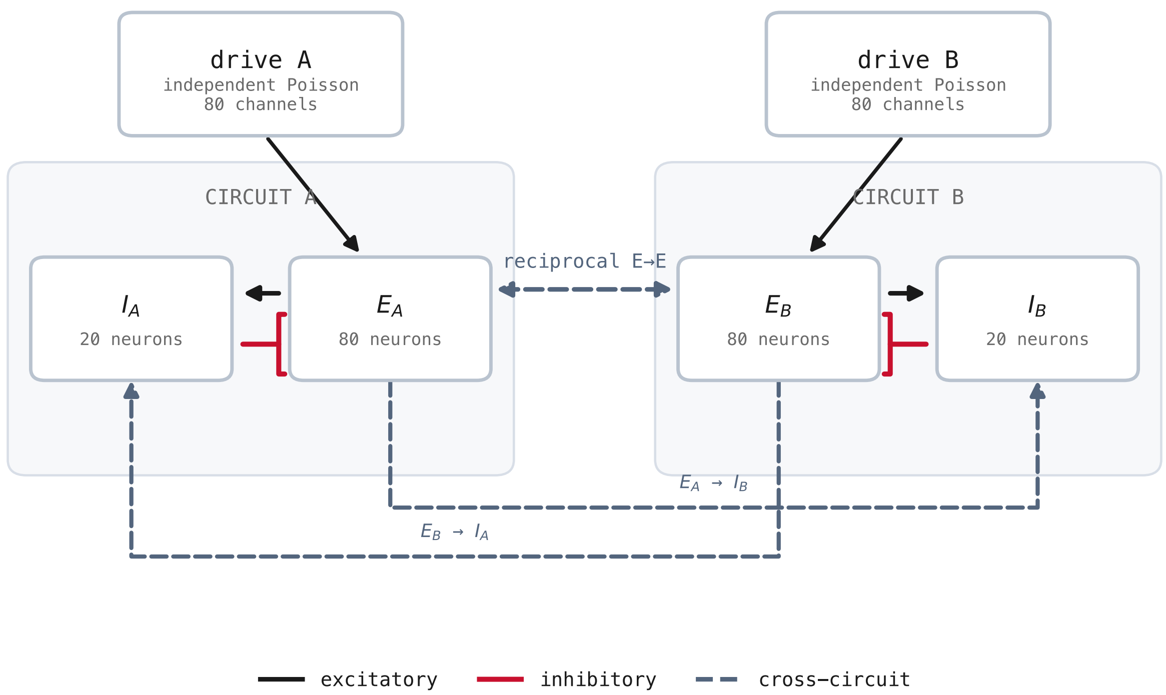

Figure 1: Authored two-circuit network. Each private Poisson input drives a local E-to-I-to-E PING loop. Dashed arrows are the four reciprocal AMPA pathways varied together by K; red bar-headed arrows are local GABA-A inhibition. The same compiled graph topology is used at every condition.

Calibration and registration

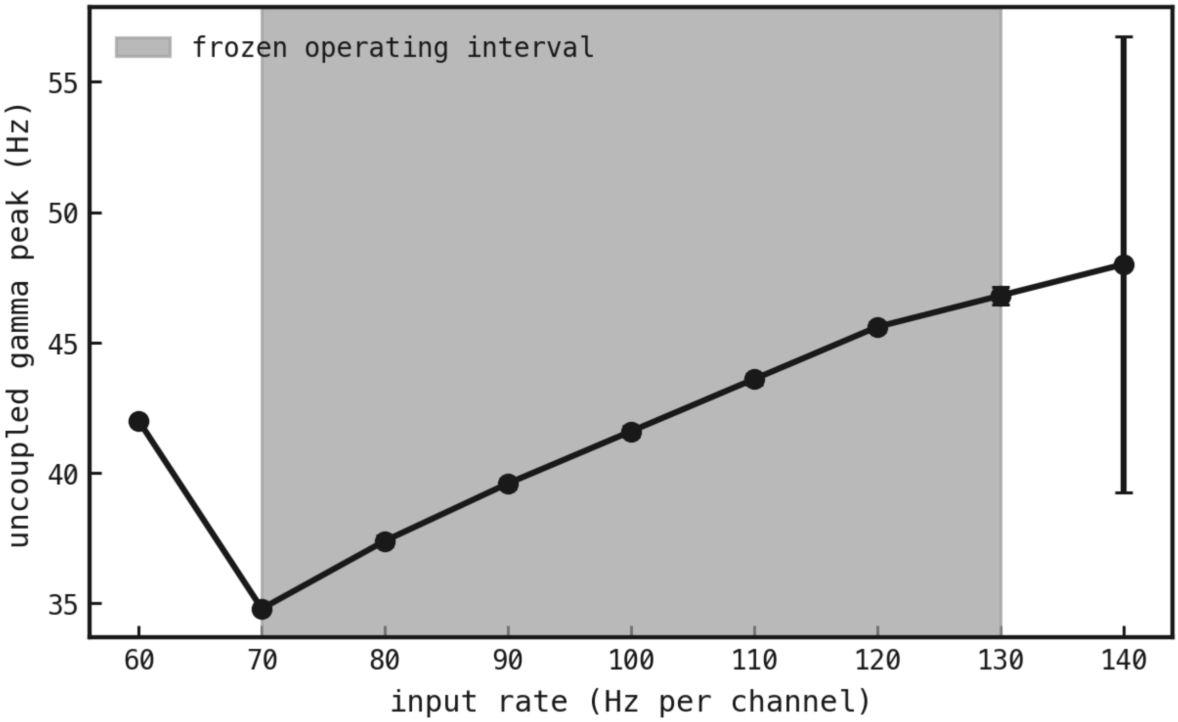

Figure 2: Uncoupled input-rate calibration. Input rate in hertz per channel is on the horizontal axis and median gamma peak frequency in hertz is on the vertical axis; bars show half the within-rate interquartile range. The shaded 70–130 Hz interval is frozen before coupled trials are inspected.

The zero-coupling calibration registers a frequency-difference tolerance of 0.8 Hz, an absolute phase-slope tolerance of 11.53 rad/s, and at most 4 complete slips. The primary-grid checksum is 8813df0395d0aeb00eb4613d28f776cc09cb4ff783694a7c7603e749c79f3c0f.

Results

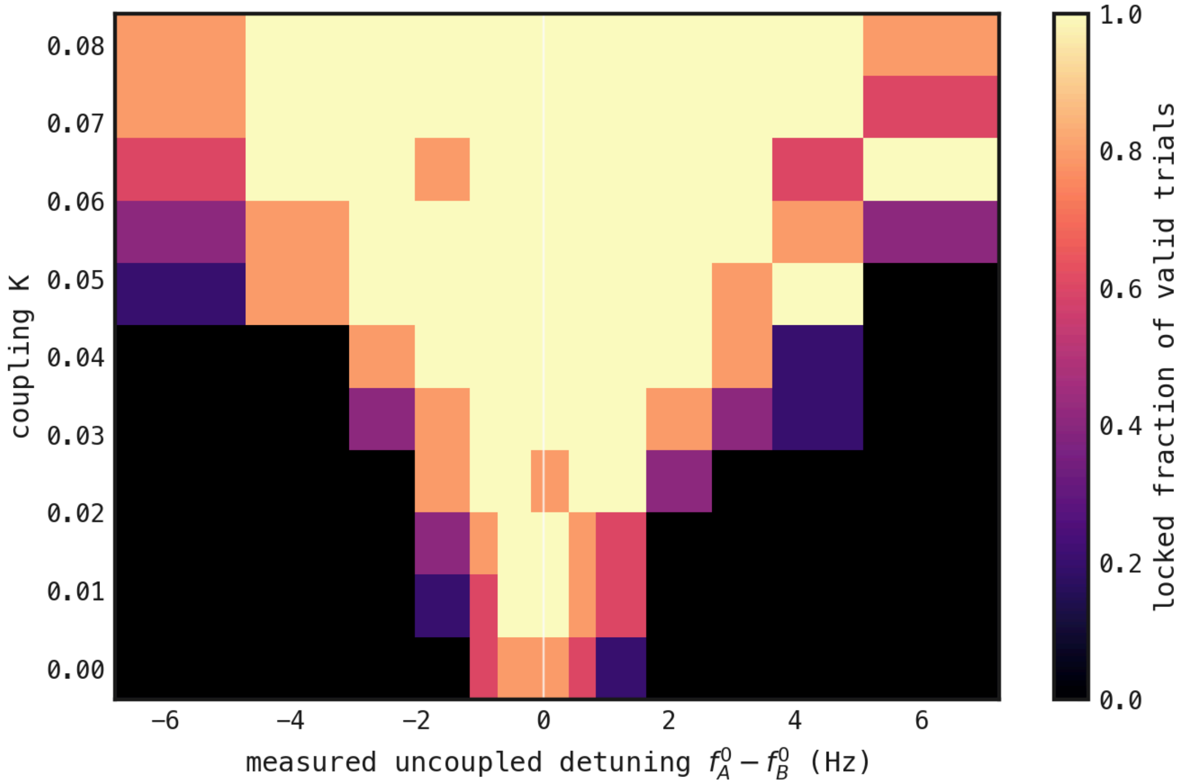

Figure 3: Primary Arnold-tongue map. Measured uncoupled detuning $\Delta f_0 = f_A^0 - f_B^0$ in hertz is on the horizontal axis and reciprocal coupling K is on the vertical axis. Colour gives the fraction of valid trials satisfying the registered frequency, phase-drift, and phase-slip tolerances. The centred locked width reaches 11.92 Hz at the largest registered coupling and widens across 4 successive steps.

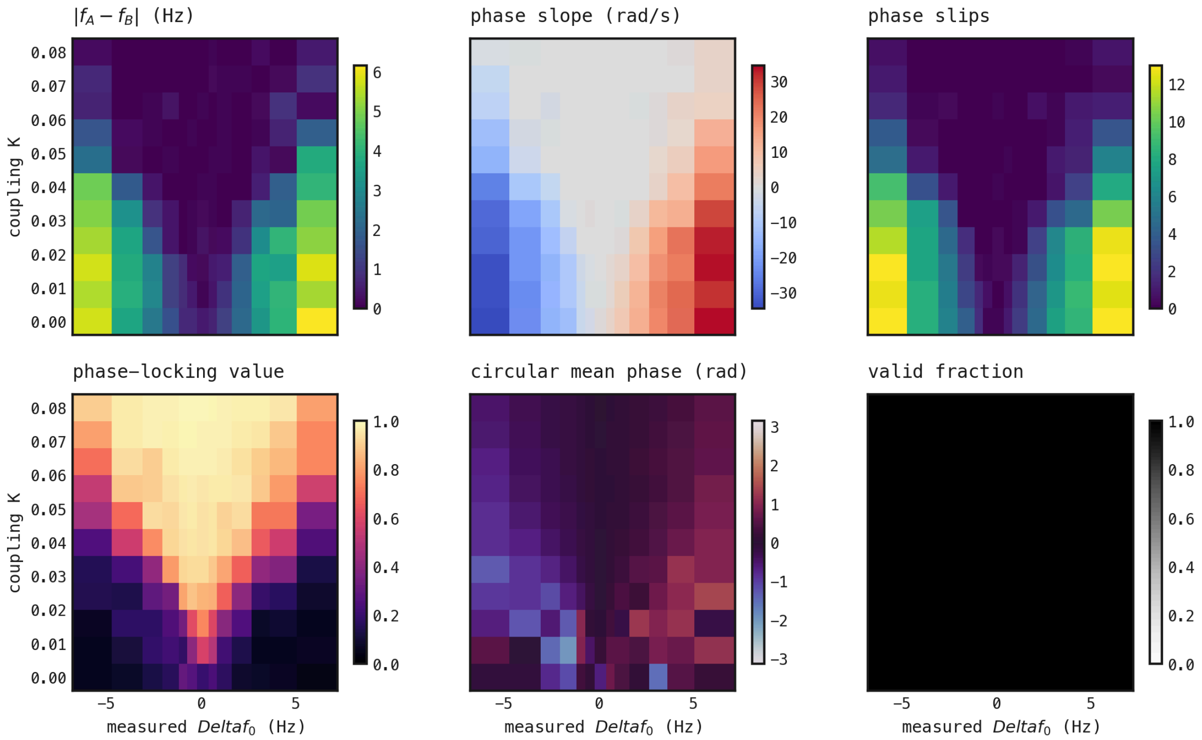

Figure 4: Supporting maps on the same measured-detuning by coupling grid. Panels show emergent absolute frequency difference in hertz, relative-phase slope in radians per second, complete phase slips, phase-locking value, circular mean phase in radians, and valid-trial fraction. Validity is 100% across the primary trials, so the tongue is not produced by silent or numerically invalid conditions.

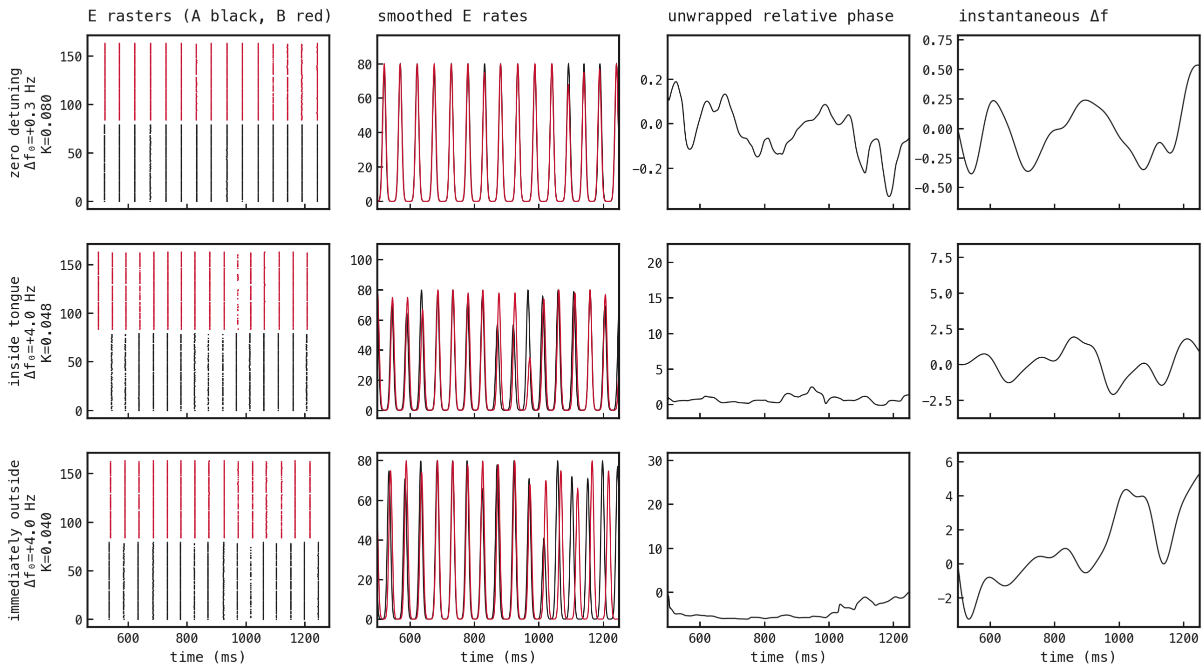

Figure 5: Predetermined representative traces. Rows show the target-zero condition at maximum coupling, the first locked +4 Hz target condition, and the same detuning at the immediately preceding coupling. Columns show both E rasters, 5 ms-smoothed E rates, unwrapped relative phase, and instantaneous frequency difference. The paired locations illustrate stable phase at the registered inside cell and drift immediately outside it.

Verdict and scale decision

The Arnold-tongue geometry passes: the locked region is centred near zero and its width increases for 4 successive coupling transitions. Trial validity also passes. The phase-lead clause fails in one qualifying cell, leaving a lead-sign agreement of 98.7%. Because the protocol requires agreement in every cell, the overall 80 E / 20 I reproduction failed.

The narrow failure occurs where measured detuning is close to the estimator's registered resolution, while the remaining qualifying cells show the expected sign. The reduced 800 E / 200 I confirmation therefore tests only target detunings −1 and +1 Hz at $K = 0$, 0.016, and 0.024, with ten seeds and a 5 s post-transient window. All 40 coupled trials are valid and locked, and all 40 have the expected phase sign. At the previously disputed negative-detuning cell, all 10 seeds have the correct sign with mean phase −0.543 rad; the mirrored positive condition gives 0.526 rad. The focused confirmation therefore passes and supports finite-size/resolution noise, rather than a systematic contradiction, as the cause of the lone 80 E / 20 I failure.

Dense retention would occupy 27.82 GB. The published exact-event and state-summary archive occupies 165.5 MB, a 168.1-fold reduction. The 143 primary cells required a median 19.03 s each locally.

References

1. Lowet, Roberts, Hadjipapas, Peter, van der Eerden & De Weerd (2015): *Input-Dependent Frequency Modulation of Cortical Gamma Oscillations Shapes Spatial Synchronization and Enables Phase Coding*. PLOS Computational Biology. doi:10.1371/journal.pcbi.1004072